

Movie Metrix

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Abstract — In today's digital age, the overwhelming abundance of available movies makes it increasingly challenging for users to find content that matches their preferences and moods, creating a demand for effective recommendation systems. "MovieMetrix" addresses this need by offering a sophisticated movie recommendation system developed using Streamlit, which utilizes advanced natural language processing techniques to deliver personalized and mood-based movie suggestions. Leveraging TF-IDF Vectorizer for text similarity and TextBlob for sentiment analysis, the application processes a comprehensive movie dataset through tokenization, stopword removal, and lemmatization using NLTK. Our system features personalized recommendations based on user input and mood-based recommendations identified through sentiment analysis, supported by a predefined mood-to-genre mapping. Additionally, MovieMetrix includes filtering options by director, actor, or genre to enhance the user discovery experience. The results demonstrate that MovieMetrix effectively tailors recommendations to user inputs and moods, providing a user-centric solution for movie discovery and highlighting its potential applications and future improvements in personalized entertainment recommendations.

Keywords — Movie Recommendation, NLP, Content-based Filtering, Mood-based Recommendation, TF-IDF, Cosine Similarity

I. INTRODUCTION

The exponential growth of digital content, particularly in the film industry, has resulted in an overwhelming number of choices for viewers. This abundance, while beneficial, often leads to decision fatigue and difficulty in selecting content that aligns with individual preferences and current moods. Consequently, there is a significant demand for intelligent recommendation systems that can navigate this vast array of options and provide tailored suggestions to users.

"MovieMetrix" is designed to address this challenge by offering a sophisticated movie recommendation system that combines advanced natural language processing (NLP) techniques with an intuitive user interface built using Streamlit. The primary objective of MovieMetrix is to enhance the movie selection

process by delivering personalized recommendations based on user inputs and mood analysis. By integrating TF-IDF Vectorizer for measuring text similarity and TextBlob for sentiment analysis, MovieMetrix can effectively understand and process user preferences and sentiments.

The system's core functionality includes two main recommendation modes: Personalized Recommendations, which generate movie suggestions based on detailed user input, and Mood-Based Recommendations, which identify the user's current mood through sentiment analysis and recommend movies that correspond to that mood. Additionally, MovieMetrix provides robust filtering options, allowing users to discover movies based on specific criteria such as director, actor, or genre.

This paper details the design, implementation, and evaluation of MovieMetrix, illustrating how it leverages NLP techniques and sentiment analysis to offer a user-centric movie recommendation experience. We discuss the preprocessing of a comprehensive movie dataset, the methods used to enhance recommendation accuracy, and the user interface design aimed at providing an engaging and immersive experience. The results demonstrate the system's effectiveness in tailoring recommendations to user inputs and moods, highlighting its potential applications and future improvements in the realm of personalized entertainment recommendations.

II. LITERATURE REVIEW

In this paper discusses the utilization of sentiment analysis in movie recommendation systems, leveraging text reviews to enhance recommendation accuracy. It highlights the application of various machine learning models like SVM for sentiment classification and references several studies on sentiment analysis and text classification techniques [1].

The paper highlights various hybrid recommendation models, which combine user interaction data, text

mining, and geographic information to enhance recommendation accuracy in social networks and tourism services, illustrating their effectiveness in providing personalized recommendations in diverse application fields [2].

This paper discusses the importance of user satisfaction in building effective recommendation systems for movies. It compares different filtering methods like content-based, collaborative, and context-based filtering. The focus is on improving accuracy and simplicity in the recommendation process. The proposed system uses collaborative filtering with nearest neighbors for better results. Overall, the goal is to enhance the accuracy and efficiency of movie recommendation systems for users [3].

The document details a personalized movie recommendation system using Content-based Filtering, leveraging user preferences and Natural Language Processing (NLP) for text analysis. It converts movie descriptions into vectors to calculate similarities, offering accurate suggestions. The system's design and testing demonstrate its reliability and future potential in various fields [4].

The content discusses various methods used in movie recommender systems, focusing on incorporating human emotions into the recommendations. It highlights three main recommendation approaches: Collaborative Filtering, Content-Based Filtering, and a Hybrid approach combining both. The text emphasizes the importance of understanding users' emotions to enhance movie recommendations. It also outlines a new hybrid recommender system framework that integrates user emotions with traditional recommendation techniques. The ultimate goal is to provide users with personalized and emotionally relevant movie suggestions [5].

It introduces a hybrid approach combining two filtering techniques, using Support Vector Machine and genetic algorithm. The goal is to improve the accuracy, quality, and scalability of movie recommendations. By merging these methods, the system aims to provide better movie suggestions based on user preferences. The document also mentions the importance of entertainment in our lives and the need for reliable recommendation systems due to the vast number of movies available online [6].

The document discusses the application of a content-based approach in creating a paper recommendation

system for digital libraries. It explains how this system uses techniques like TF-IDF and cosine similarity to suggest relevant research papers to users based on their queries. The paper highlights the importance of recommender systems in overcoming information overload and presents an algorithm for generating recommendations. It also mentions the potential future work of combining collaborative and content-based filtering techniques for improved recommendations [7].

III. METHODOLOGY

The implementation of a movie recommendation system based on Content-Based Filtering and Emotion i.e Mood-Based Filtering begin by importing the necessary libraries. Starting with Python library, Pandas which is used for loading the .csv file of the raw text data that will be used in training and testing the model. Then import sklearn for importing TfidfVectorizer and cosine_similarity. Importing textblob to detect the sentiment polarity of the user prompt. Nltk is used for applying stopwords, tokenization and lemmatization.

The next important step is data cleaning and making it ready to pass into the model. Removing rows containing NULL values, replacing NULL values with NA, removing all special characters and extra spaces, etc. The step continues by tokenizing the corpus to documents (sentences) and to words, lowering all alphabets, and then removing all stop words in the user prompt and description of movies. Then, using lemmatizer.lemmatize(), lemmatize, i.e., convert all the words of the corpus to their original form. For example, lemmatizer.lemmatize('runs') will convert the word to 'run' i.e. to its base form.

Apply all preprocessing to all text elements of the data set.

The Tfidf (Term Frequency and Inverse Document Frequency) approach converts the text to vectors so that we could apply the model on it. TF is calculated for every word and every sentence and IDF is calculated for every word appearing in every document.

$$TF = (\text{Number of repetition of a word in a sentence}) / (\text{Number of words in a sentence})$$

$$IDF = \log_e ((\text{number of sentences}) / (\text{Number of sentences containing the word}))$$

$TF\text{-}IDF(t,d)=TF(t,d) \times IDF(t)$, where t is the term and d is the document or sentence.

TF-IDF gives importance to the rare words appearing in a particular sentence. It makes the words to have less importance which occur in as many documents. Say, if a word 'fun' appears in almost all documents, the TF-IDF works in such a way that it makes its importance in the vectorizer negligible. Similarly, if a word occurs rarely in any sentence, it gives more importance.

Now, according to the user input, Personalized recommendation or Mood based recommendation), the model switches to the function that performs the recommendation.

A) Personalized Recommendation System: This takes the user prompt as the parameter in the function. This first preprocesses the user input. Then the Tfidf vectorizer uses `fit_transform()` to vectorize the processed text of the data set. Then, use `transform()` on the user input to convert it to vectors. The task of utmost importance after vectorizing is to find the function that finds the similarities between the input and data set. As a result of the literature review, we can conclude to use cosine similarity. It is used to measure the similarity between the user's input and the movie descriptions.

Cosine Similarity is the dot product of the vectors divided by the product of their lengths., or simply, $\text{Cosine Similarity} = 1 - \cos(x)$, where x is the angle made by the test and train data points from the origin.

When cosine similarity equals to 0, it is a strong indication that the test and train points are extremely similar in terms of their features.

The work is not yet done. We now need to sort the recommendations according to their similarities and display the top recommendations.

B) Emotion Mood-Based Recommendation System: The user's mood is a key to recommending movies. It first processes the user input, applies stop words, tokenizes, and lemmatizes. The detection of mood is done using Textblob. Finding the mood inculcates the use of `blob.sentiment.polarity` to calculate the sentiment score of the user input text. The polarity suggests the mood of the user. The polarity ranges from -1 to 1, where sentiment score ≥ 0.5 is 'happy' sentiment score ≤ -0.5 is 'sad'

sentiment score < 0 is 'negative'
sentiment score > 0 is 'positive'
sentiment score = 0 is 'neutral'

Now, the mood is set to the corresponding mood. Then, this mood is mapped to preferred genres that are further used to filter the movie data points. Then, movies are filtered based on the tags related to the mood. Now, this becomes the 'Mood Processed Text'. This text is then vectorized, and the recommendations are sorted based on their cosine similarities. This is how the whole process works.

Note: Stemming is not applied because it may reduce precision and convert important text to meaningless words, losing context in the search.

C) Filtering: Apart from recommendation, the app allows the user to filter movies based on multiple factors, like actors, directors, and genre. It inculcates a normal filtering option where the app scans the corpus and returns all the data points where the input filter matches the document in the corpus.

Streamlit:

The movie recommendation system relies heavily on streamlit, which offers an interactive interface that allows users to input their preferences or moods and obtain personalized movie recommendations. Every suggested film is presented by Streamlit with a wealth of information, such as the poster, title, genre, languages, director, writer, actors, synopsis, similarity score, IMDb rating, and a Netflix watch link. With the use of background pictures, text styling, and layout modifications, custom CSS styling improves the visual presentation and produces an engaging user experience.

Moreover, Streamlit allows recommendation algorithms to be executed in real time. Users receive instant feedback and increased user satisfaction when the app processes their input, computes cosine similarity, and displays the top recommended movies.

To sum up, the recommendation system is transformed from a purely mathematical model into a fully interactive web application by using Streamlit. Through this interface, users may easily access and use the recommendation service, knowing that the system is reliable, accessible, and easy to use.

IV. RESULT AND DISCUSSION

Homepage:

The homepage of our application already features personalized movie recommendations and a navbar with options to select personalized movie recommendations, mood-based recommendations, and a filter movies option.

Navbar:

Navbar have a feature of selecting the option of personalized movie recommendation, mood based recommendation and filter movies.

Personalized Movie Recommendation:

The homepage of the Movie Recommendation System displays a personalized movie recommendation feature. Users can enter their preferences in a text box, such as "I want to watch a Hindi movie in which wrestlers fight for a gold medal in sports," and then click the "Get Recommendations" button to receive tailored movie suggestions. The top recommendation displayed is the movie "Dangal."

Mood-based Recommendation:

The Movie Recommendation System homepage also features a mood-based recommendation option. Users can describe their current mood in a text box, such as "I am feeling sad, lonely, and tensed," and then click the "Get Recommendations" button to receive relevant movie suggestions. The top recommendation shown is the movie "Blaumacher," categorized under Comedy and Drama.

Filter Movies:

It gives the user the option of selecting director, actor and genres and the user can filter the movies.

V. FUTURE SCOPE

The future scope of "MovieMetrix" includes several key advancements to enhance its capabilities. Leveraging more powerful vectorization techniques, such as word embeddings and contextual embeddings like BERT, will improve semantic understanding of movie descriptions and user inputs. Implementing collaborative filtering and hybrid recommendation systems will combine user behavior patterns and movie characteristics for more accurate suggestions. Integrating Large Language Models (LLMs) like GPT variants will refine insights into user preferences and movie attributes. Expanding cross-platform integration with various streaming services and enhancing user profiling techniques will further personalize the user experience. Adding diverse

language support, social features for sharing recommendations, and advanced content analysis techniques will increase accessibility and engagement. Developing a mobile application will provide convenient, on-the-go access. These enhancements will enable MovieMetrix to deliver highly personalized and relevant movie suggestions to a broader audience.

VI. CONCLUSION

"MovieMetrix" represents a significant advancement in the field of personalized movie recommendation systems, addressing the challenge of overwhelming content choices in the digital age. By leveraging advanced natural language processing techniques, such as TF-IDF Vectorizer and TextBlob sentiment analysis, along with sophisticated preprocessing of movie data, the system effectively tailors recommendations to individual user preferences and moods. The implementation of personalized and mood-based recommendation modes, combined with robust filtering options, enhances the user experience by providing relevant and engaging movie suggestions. Future enhancements, including the integration of more powerful vectorization methods, collaborative filtering, hybrid recommendation systems, and advanced user profiling, will further refine the accuracy and personalization of recommendations. As MovieMetrix continues to evolve, it holds the potential to become a comprehensive and highly sophisticated tool for personalized entertainment discovery, catering to a diverse and global audience.

VII. REFERENCES

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